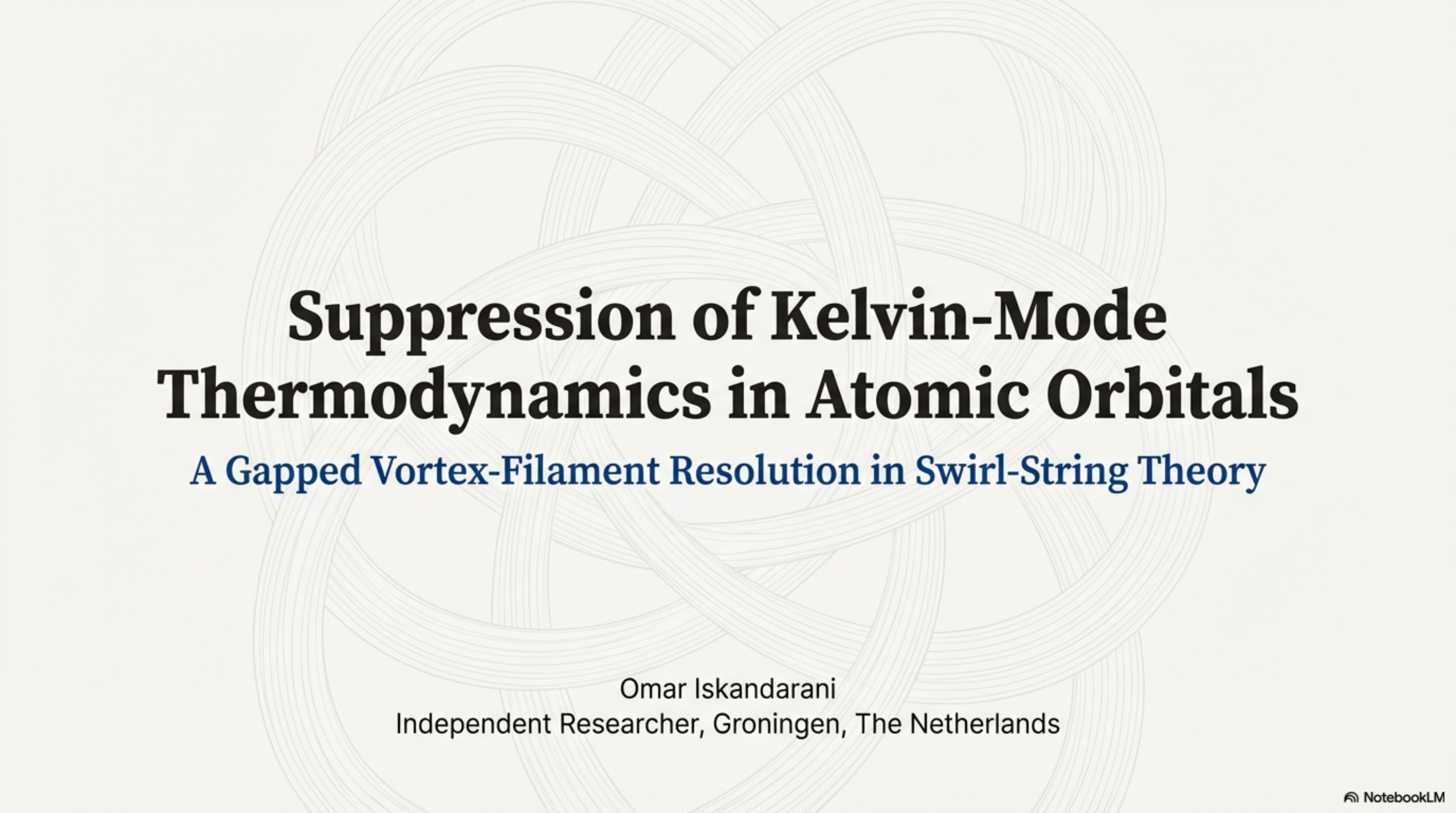


The background of the slide features a complex, repeating pattern of interlocking circles or rings, rendered in a light gray color. This pattern creates a sense of depth and complexity, reminiscent of a woven fabric or a mathematical structure like a braid.

Suppression of Kelvin-Mode Thermodynamics in Atomic Orbitals

A Gapped Vortex-Filament Resolution in Swirl-String Theory

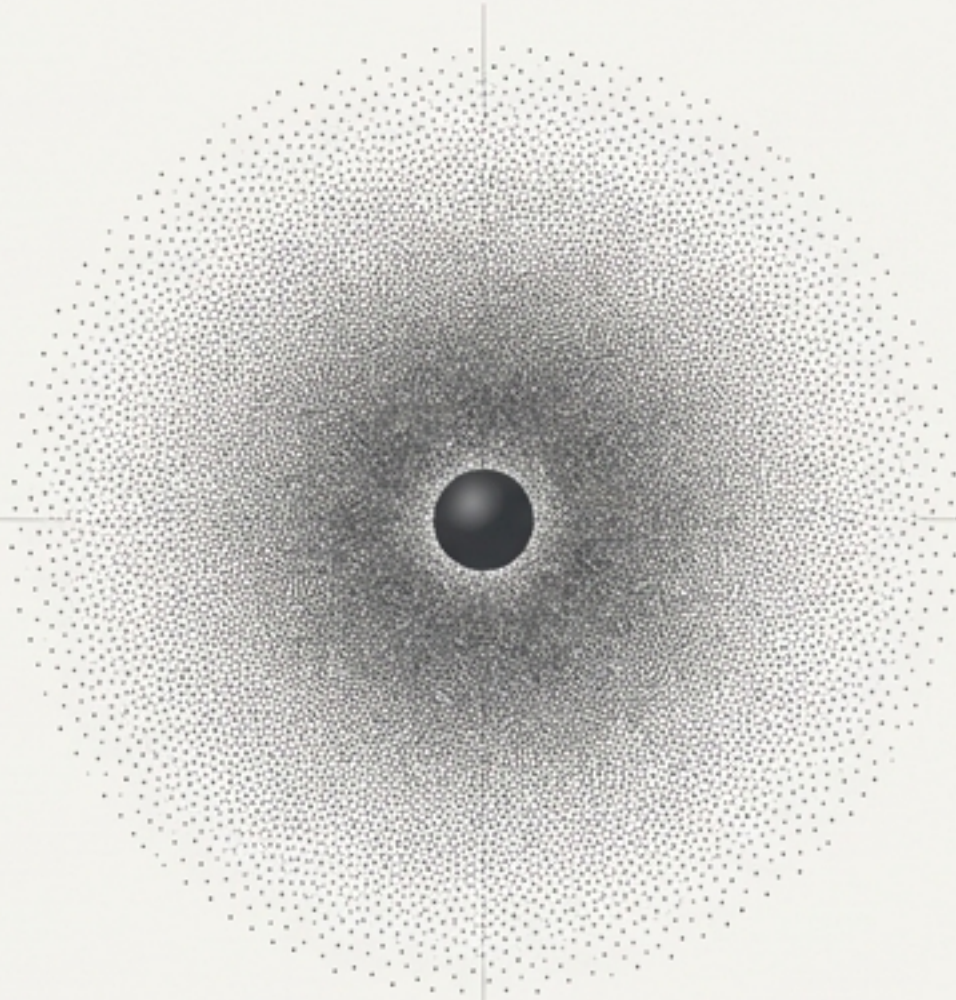
Omar Iskandarani

Independent Researcher, Groningen, The Netherlands

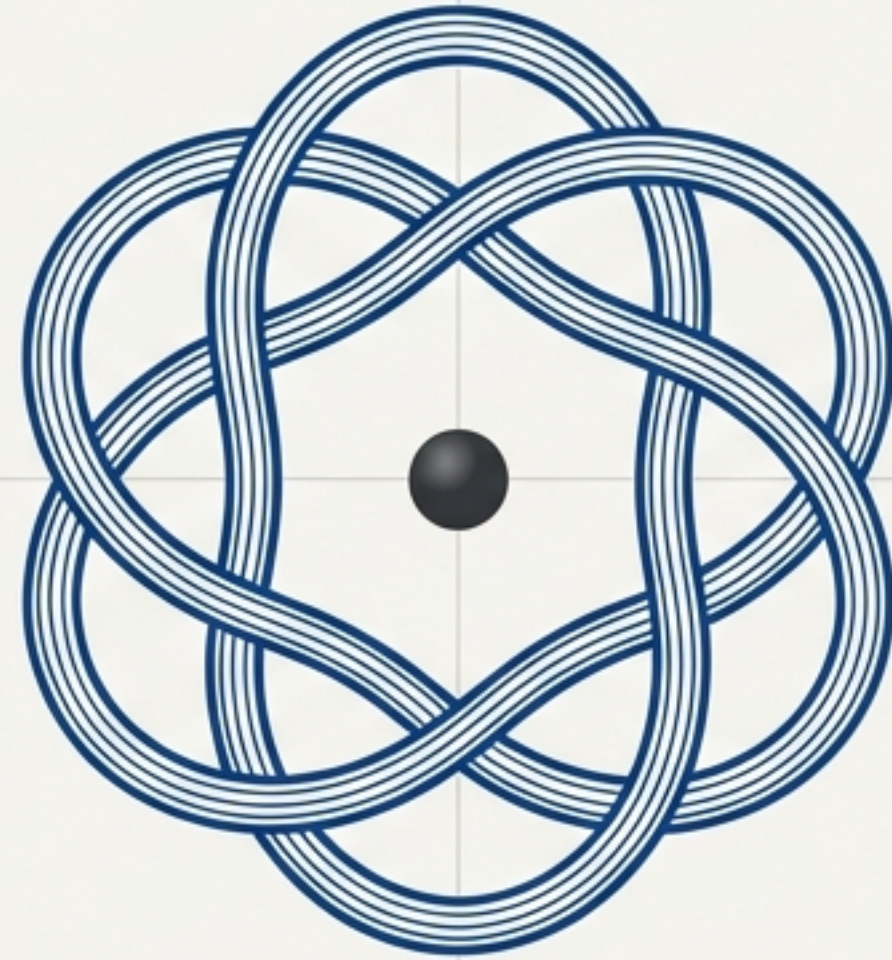
A New Physical Picture: Particles as Knotted Vortex Filaments

Swirl-String Theory (SST) posits that elementary particles are not point-like probability clouds, but stable, knotted vortex filaments embedded in a physical, incompressible condensate.

Standard Quantum Mechanics



Swirl-String Theory



This model revives the vortex-atom theories of Kelvin and Helmholtz, grounding quantum phenomena in tangible hydrodynamic principles.

Quantum Mechanics Emerges from Hydrodynamic Principles

In SST, the defining features of atomic physics are not postulated but derived. The discrete energy levels of the hydrogen atom arise from hydrodynamic force-balance conditions.

The system seeks to minimize a free-energy functional, $F[\psi]$.



This functional includes terms for the filament's kinetic energy ($|\nabla\psi|^2$) and a potential energy $V_{\text{SST}}(\mathbf{r})$ arising from hydrodynamic pressure gradients, which naturally yields a $-1/r$ potential.



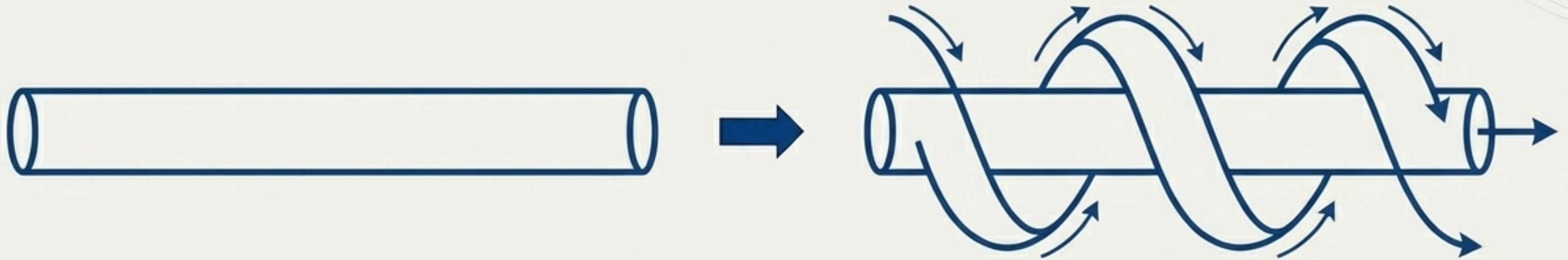
Varying this functional to find the stable equilibrium state directly yields the stationary **Schrödinger equation:**

$$-\frac{\hbar^2}{2m_e}\nabla^2\psi + V_{\text{SST}}(\mathbf{r})\psi = E\psi$$

The Schrödinger equation is thus an effective equation of state for the vortex filament's configuration, not a fundamental axiom.

An Inherent Feature of Filaments: Kelvin Waves

Any real vortex filament can support helical excitations that propagate along its length. These are known as Kelvin waves. They are a natural, unavoidable consequence of the model.



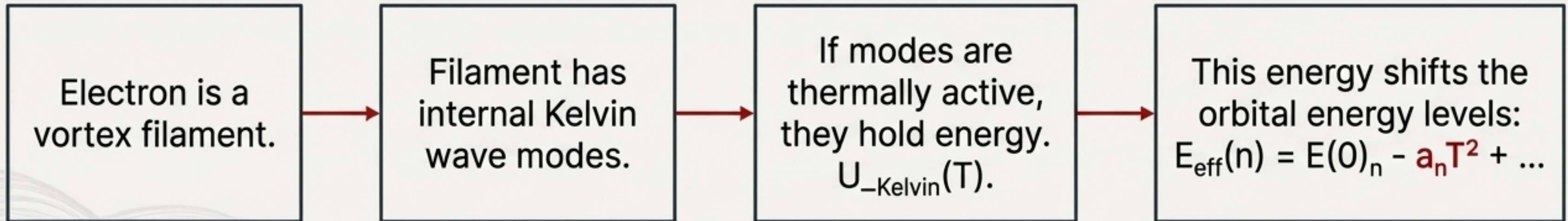
Dispersion Relation

$$\omega(k) \simeq \frac{\Gamma}{4\pi} k^2 \left[\ln(1/|k|\xi) + C_0 \right]$$

Implication: For a closed filament in an atomic orbital of length L_n , the allowed modes are quantized ($k_m = 2\pi m/L_n$), meaning the electron filament possesses an internal spectrum of vibrational modes.

The Thermodynamic Threat to Atomic Spectra

If these internal Kelvin modes can be thermally excited, they would contribute to the atom's total internal energy. This would introduce temperature-dependent corrections to the precisely measured atomic energy levels.



The Stakes: The hydrogen spectrum is one of the most precisely measured systems in physics. Any deviation from the standard $1/n^2$ energy structure must be incredibly small.

A Discrepancy of 20+ Orders of Magnitude

Naive SST Prediction

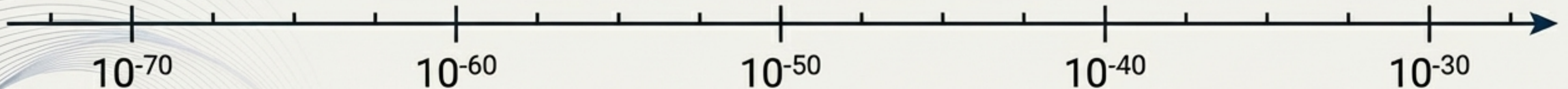
A straightforward elastic estimate of the **thermal correction coefficient** a_n using canonical SST parameters.

$$a_n (\text{naive}) \sim 10^{-39} \text{ J/K}^2$$

Experimental Reality

The **upper bound on any such correction coefficient**, derived from high-precision spectroscopy of the hydrogen atom.

$$a_n (\text{exp}) \lesssim 10^{-62} \text{ J/K}^2$$

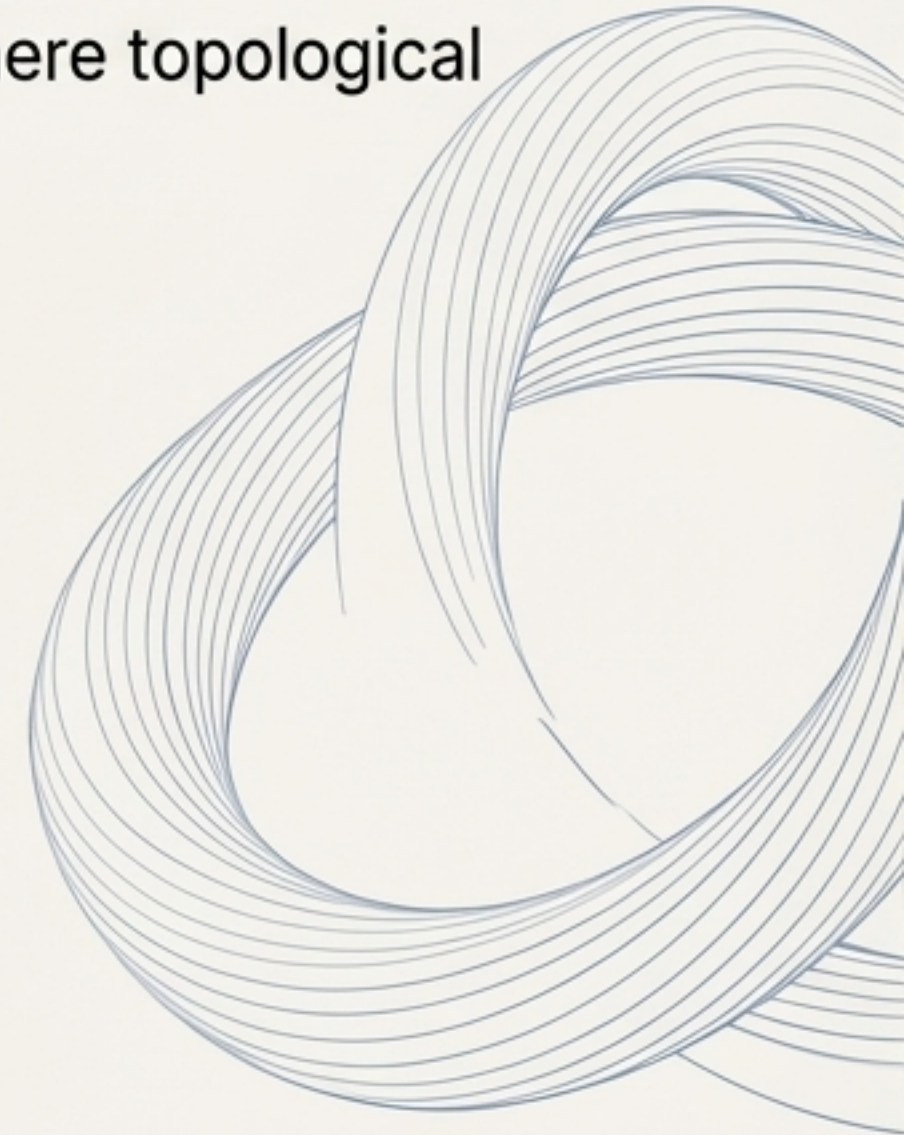
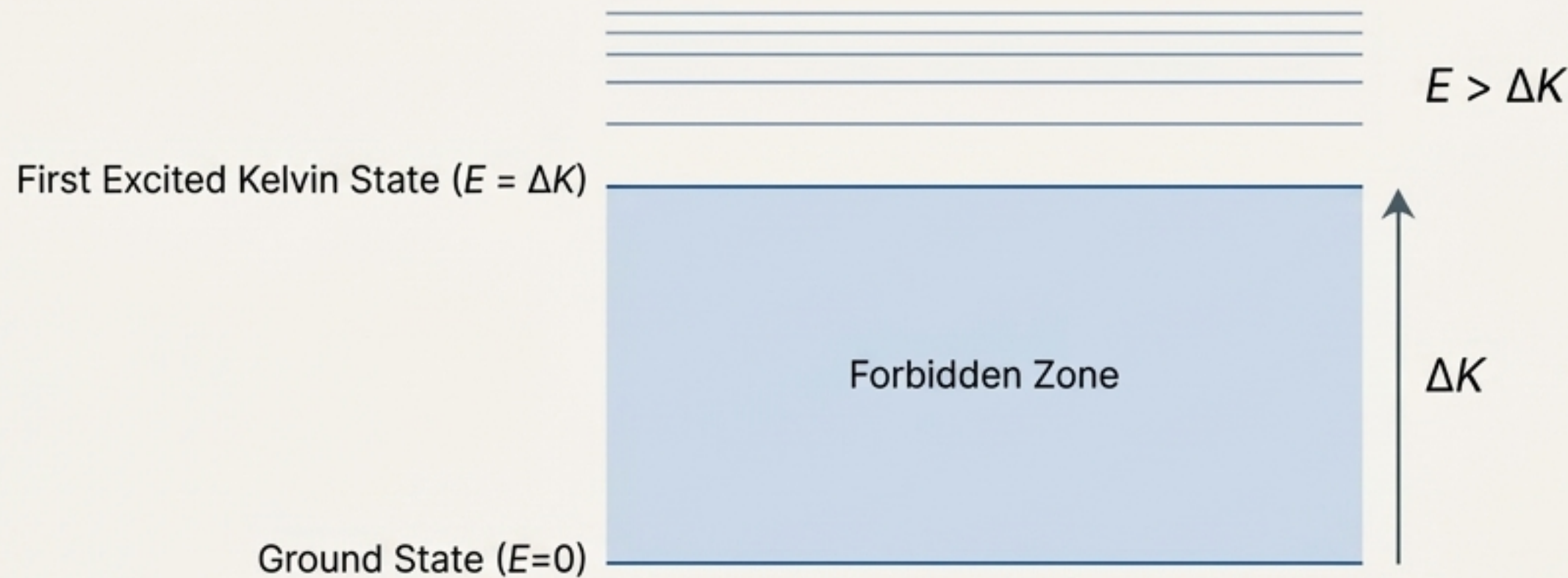


Without a suppression mechanism, naive Kelvin thermodynamics would catastrophically distort atomic energy levels. The theory faces a fatal flaw.

The Resolution: A Topologically Induced Excitation Gap

The Proposal: The Kelvin spectrum of the electron filament is not continuous from zero energy. It is gapped. There exists a minimum energy, ΔK , required to excite the lowest possible Kelvin mode.

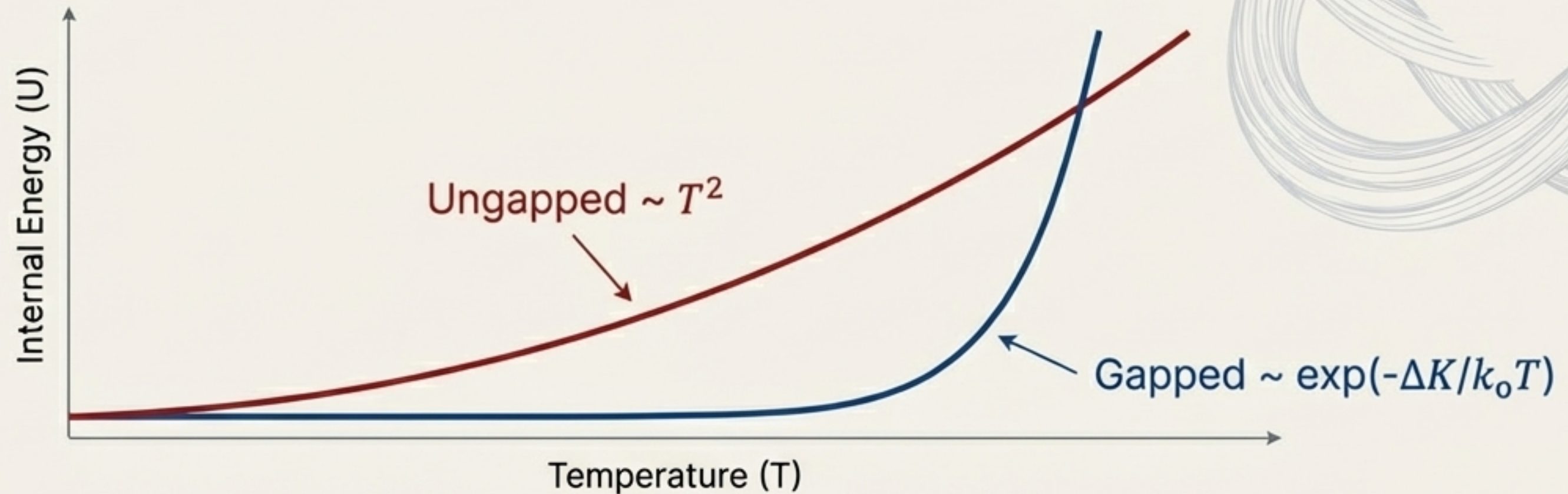
Supporting Rationale: Such gaps are natural for complex, knotted filaments, where topological constraints, curvature, and torsion prevent arbitrarily small deformations.



The Gapped Hamiltonian: The energy of any mode m in orbital n must satisfy $E_{m,n} \geq \Delta K$.

The Power of the Gap: Exponential Suppression

The Mechanism: An energy gap ΔK fundamentally changes the low-temperature thermodynamics. At temperatures where $k_o T \ll \Delta K$, there is insufficient thermal energy to excite the mode.



The internal energy U of a gapped mode is dominated by an exponential factor:

$$U \approx \Delta K \cdot e^{-\Delta K/k_o T}$$

This exponential suppression replaces the dangerous polynomial behavior, effectively “freezing out” the Kelvin modes at atomic energy scales.

Determining the Required Scale of the Energy Gap

Step 1: The Constraint

To reconcile theory with experiment, the exponential suppression must be strong enough to reduce the predicted a_n by over 20 orders of magnitude. This leads to a required thermodynamic factor:

$$\frac{\Delta K}{k_o T_{\text{eff}}} \gtrsim 60$$

Step 2: Estimating the Effective Temperature

We take a conservative upper bound for the effective microphysical temperature (T_{eff}) an electron orbital might experience.

$$T_{\text{eff}} \lesssim 10^5 \text{ K}$$

Step 3: The Resulting Gap

Combining these constraints yields the required minimum energy for the Kelvin gap.

$$\Delta K \gtrsim 500 \text{ eV}$$

A Kelvin excitation gap on the order of hundreds of electron-volts is sufficient to render the modes thermodynamically inert in atoms.

A Perfectly Scaled Solution



- 1. $\Delta K \gg 10$ eV: The gap is enormous compared to atomic binding energies. This is *why* the **Kelvin modes** are completely frozen in ordinary atomic physics.
- 2. $\Delta K \ll 511$ keV: The gap is a tiny fraction of the electron's total rest energy. This makes it a **plausible** internal structural property of the electron filament, not a radical change to its fundamental mass.

A Deeper Foundation for the Schrödinger Equation

With Kelvin modes frozen and suppressed, the only relevant dynamics are the overall configuration of the electron filament.

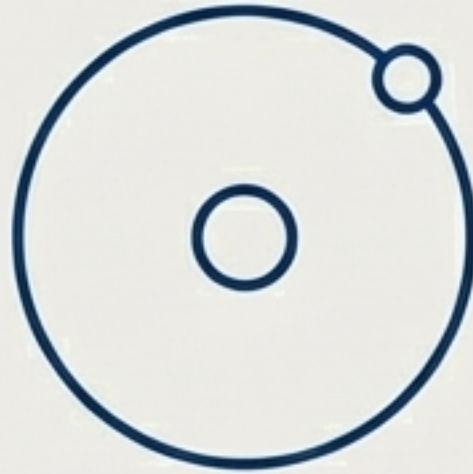
The SST free-energy functional $F[\psi]$, which accounts for the filament's hydrodynamic energy, becomes the sole determinant of the system's state.

Minimizing this functional $F[\psi]$ is mathematically equivalent to solving the stationary Schrödinger equation.

“Quantum mechanics appears as a low-energy, Kelvin-frozen limit of vortex-filament thermodynamics.” — This provides a concrete hydrodynamic realization of the quantum-thermodynamic correspondence identified by Abe and Okuyama.

A Clear Separation of Physical Regimes

Low-Energy / Low-Acceleration Regime



State: Kelvin modes are FROZEN ($k_o T \ll \Delta K$).

Governing Physics: The Schrödinger equation is the effective equation of state.

Description: This is the domain of all ordinary atomic and condensed matter physics. The internal structure of the electron is inert.

High-Energy / High-Acceleration Regime



State: Kelvin modes ACTIVATE ($E \gtrsim \Delta K$).

Governing Physics: Internal filament dynamics become relevant.

Description: This is the domain of novel SST phenomenology, accessible in high-energy scattering, extreme acceleration (Swirl-Unruh regimes), and cosmology.

New Questions on the Horizon

The resolution of the thermodynamic constraint opens a new landscape of theoretical inquiry. Key questions for future work include:

- 🔗 **Knot Topology:** Can the magnitude of the Kelvin gap (ΔK) be derived directly from the specific knot topology of the electron filament?
- 🔗 **Lepton Generations:** Does the structure of Kelvin excitation gaps correlate with the existence of the three generations of leptons (electron, muon, tau)?
- 🔗 **New Phenomenology:** What are the specific experimental signatures of activated Kelvin modes in:
 - keV-MeV scattering processes?
 - Astrophysical environments like neutron stars?
 - Ultra-high acceleration regimes?

Appendix: Key Mathematical Formulations

Section 1: Kelvin Wave Dispersion

$$\omega(\mathbf{k}) \simeq \frac{\Gamma}{4\pi} k^2 \left[\ln\left(\frac{1}{|\mathbf{k}|\xi}\right) + C_0 \right]$$

Low-Temperature Internal Energy (Single Mode)

$$U = \frac{\Delta K}{e^{\beta\Delta K} - 1} \approx \Delta K e^{-\frac{\Delta K}{k_o T}} \text{ for } k_o T \ll \Delta K$$

Section 2: Gapped Kelvin Hamiltonian

$$H_K^{(n)} = \sum_m \left[(\Delta K + \delta E_{m,n}) b_{mn}^\dagger b_{mn} + \frac{1}{2} (\Delta K + \delta E_{m,n}) \right]$$

Key Parameters & Constraints

- Experimental Limit: $a_n \lesssim 10^{-62} \text{ J/K}^2$
- Required Gap Condition: $\frac{\Delta K}{k_o T_{\text{eff}}} \gtrsim 60$
- Calculated Gap: $\Delta K \gtrsim 500 \text{ eV}$

References

1. H. Helmholtz, 'On Integrals of the Hydrodynamical Equations Which Express Vortex Motion,' J. Reine Angew. Math. 55, 25-55 (1858).
2. W. Thomson (Lord Kelvin), 'On Vortex Atoms,' Proc. R. Soc. Edinburgh 6, 94-105 (1867).
3. L. Onsager, 'Statistical Hydrodynamics,' Nuovo Cimento Suppl. 6, 279-287 (1949).
4. R. P. Feynman, 'Application of Quantum Mechanics to Liquid Helium,' in Progress in Low Temperature Physics, Vol. 1, North-Holland (1955).
5. W. Thomson (Lord Kelvin), 'Vibrations of a Columnar Vortex,' Philosophical Magazine 10, 155-168 (1880).
6. C. F. Barenghi, L. Skrbek, and K. R. Sreenivasan, 'Introduction to Quantum Turbulence,' Proc. Natl. Acad. Sci. USA 111, 4647-4652 (2014).
7. R. L. Ricca, 'The Contributions of Da Rios and Levi-Civita to Asymptotic Potential Theory and Vortex Filament Dynamics,' Fluid Dynamics Research 18, 245-268 (1996).
8. R. K. Pathria and P. D. Beale, Statistical Mechanics, 3rd ed., Elsevier (2011).
9. S. Abe and S. Okuyama, 'Similarity between Quantum Mechanics and Thermodynamics,' Phys. Rev. E 83, 021121 (2011).
10. W. G. Unruh, 'Notes on Black-Hole Evaporation,' Phys. Rev. D 14, 870-892 (1976).